Lick Kast Polarimetry — Exposure Time & S/N Plan

UV-excess QSO candidates (34 targets with g photometry)

Methodology (K. Leighly notes 2026-07; Chromey, "To Measure the Sky", sec. 9.5.4)

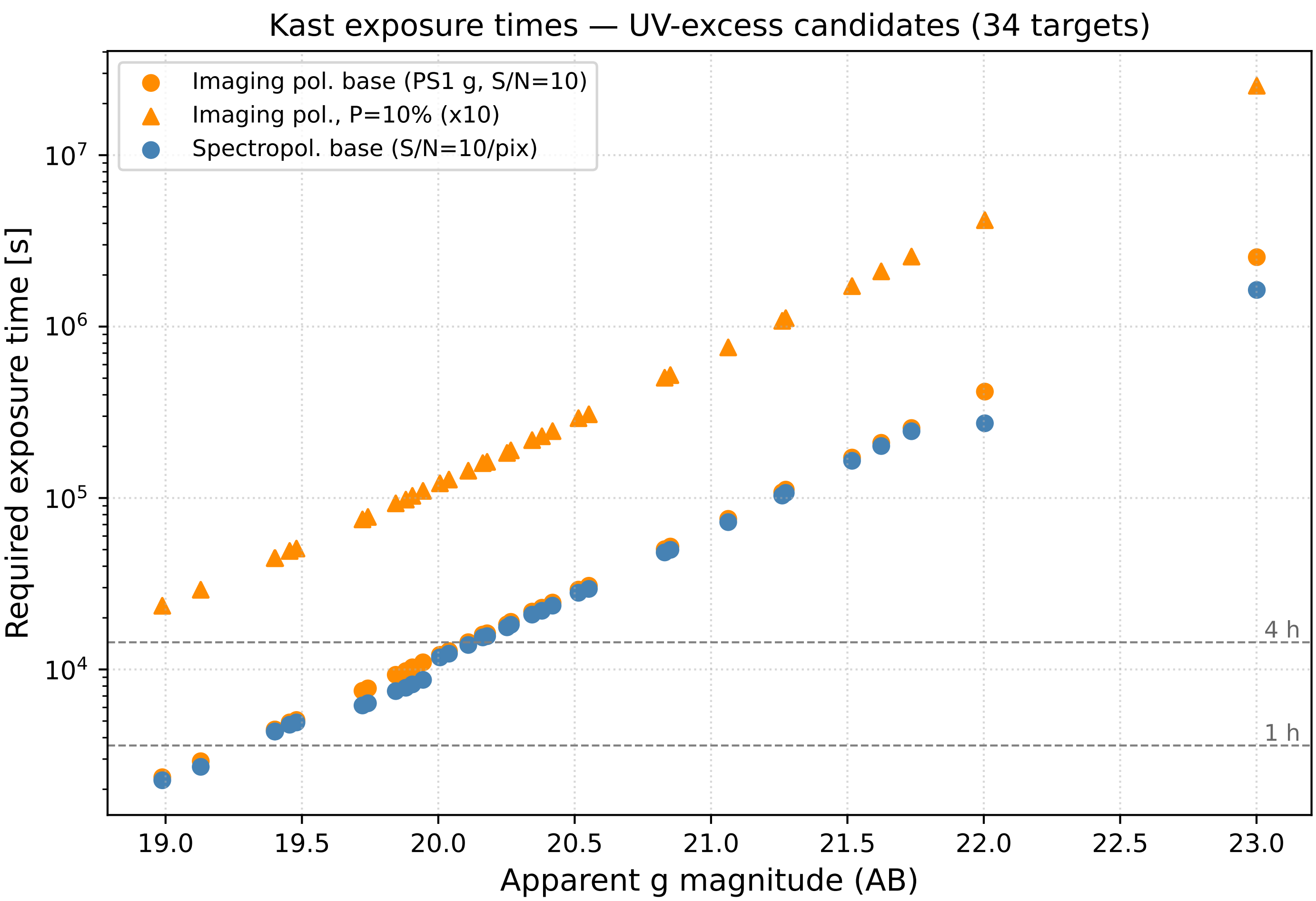
1. Each target is run through the Kast ETC (https://etc.ucolick.org/web_s2n/kast) with a flat template and its PS1 g-band AB magnitude + 0.752 mag: the polarization optics pass half the flux (m_pol = m + 2.5 log 2).
2. ETC settings, identical for all targets: dichroic d55, slit 2.0 arcsec, binning 1x1 pixels, grism 600/4310, grating 600/7500, seeing 1.5", airmass 1.1, reference exposure 900 s.
3. The ETC returns per-pixel object counts, sky counts, and read noise vs. wavelength. Spectropolarimetry uses the per-pixel S/N of the median pixel in 4000-5500 A; imaging polarimetry integrates the counts over the PS1 g profile (synphot) and applies the CCD equation S/N = Nqso / sqrt(Nqso + Nsky + Noise).
4. The exposure time reaching the required S/N is the exact solution of the CCD equation (Chromey eq. 9.77). Required S/N = 10 (advisor: "ideally >10"); the 5 floor ("definitely >5") is tabulated in kast_obs_plan.csv.
5. Polarimetry penalty: to be sensitive to polarization fraction P, the normal exposure time is multiplied by 1/P (rough rule per notes) – x10 for P = 10%, x100 for P = 1%. Note P is NOT the scattered fraction: scattering with unfavorable geometry can cancel to zero net polarization.

Sample summary at S/N = 10, imaging polarimetry (PS1 g)

P = 10%: 0 targets within 1 h, 0 within 2 h, 0 within 4 h
P = 1%: 0 targets within 2 h, 0 within 4 h, 0 within 8 h

Spectropolarimetry at S/N = 10/pixel is feasible unscaled for the brightest targets (2 within 1 h) but no target fits 4 h once the 1/P factor is applied – imaging polarimetry first, spectropolarimetry only for bright detections, is the natural protocol.

Caveats: g magnitudes are observed-frame, not corrected for Galactic extinction; the 1/P rule is approximate (a 3-sigma detection criterion via sigma_P = sqrt(2)/(S/N) scales as 1/P^2) – confirm convention before the proposal; several candidates lie at z < 0.5 (W2M rows) – vet before targeting.



Kast exposure times — UV-excess candidates (34 targets)
Imaging pol. base (PS1 g, S/N=10)
Imaging pol., P=10% (x10)
Spectropol. base (S/N=10/pix)
Required exposure time [s]
Apparent g magnitude (AB)
4 h
1 h

Per-target exposure times at S/N = 10 (brightest first; imaging = PS1 g band)

Target	g (AB)	ETC mag	Img base	Img P=10%	Img P=1%	N exp (10%)	Spec base	Spec P=10%
J204626.10+002337.6	18.99	19.74	39.2 m	6.5 h	65.3 h	14	37.6 m	6.3 h
J150746.11+093731.2	19.13	19.88	48.6 m	8.1 h	81.0 h	17	45.0 m	7.5 h
TARGETID_39628512285426154	19.40	20.15	74.3 m	12.4 h	123.9 h	25	72.4 m	12.1 h
TARGETID_39633384095352378	19.40	20.15	74.3 m	12.4 h	123.9 h	25	72.4 m	12.1 h
J150356.69+125914.3	19.45	20.21	81.8 m	13.6 h	136.4 h	28	79.3 m	13.2 h
J144809.53+054019.2	19.48	20.23	84.5 m	14.1 h	140.9 h	29	81.8 m	13.6 h
TARGETID_39628516710418961	19.72	20.47	2.1 h	20.8 h	208.4 h	42	1.7 h	17.1 h
J150357.14+041856.3	19.74	20.49	2.2 h	21.5 h	215.4 h	44	1.8 h	17.6 h
J141354.73+110047.5r	19.84	20.60	2.6 h	25.9 h	258.5 h	52	2.1 h	20.8 h
J152259.51+032347.6r	19.88	20.63	2.7 h	27.2 h	271.8 h	55	2.2 h	21.7 h
J143027.14+152514.8	19.91	20.66	2.9 h	28.6 h	285.8 h	58	2.3 h	22.7 h
J134039.67+051419.7	19.94	20.70	3.1 h	30.6 h	305.7 h	62	2.4 h	24.2 h
J150152.94+020604.9	20.01	20.76	3.4 h	33.8 h	338.2 h	68	3.3 h	32.7 h
J115449.31+112705.8	20.04	20.79	3.6 h	35.6 h	355.8 h	72	3.4 h	34.3 h
TARGETID_39633426973720698	20.11	20.86	4.0 h	40.1 h	400.6 h	81	3.9 h	38.6 h
J110018.80+080455.3	20.16	20.92	4.4 h	44.4 h	443.7 h	89	4.3 h	42.7 h
J132210.17+084232.9	20.18	20.93	4.5 h	45.1 h	451.4 h	91	4.3 h	43.4 h
TARGETID_39633267170738647	20.25	21.00	5.1 h	50.9 h	508.9 h	102	4.9 h	48.9 h
J150013.42+123645.2	20.27	21.02	5.3 h	52.7 h	526.6 h	106	5.1 h	50.6 h
TARGETID_39633136459448983	20.34	21.10	6.0 h	60.4 h	604.4 h	121	5.8 h	58.0 h
TARGETID_39633149587623462	20.38	21.13	6.4 h	63.7 h	636.5 h	128	6.1 h	61.1 h
TARGETID_39633325387681000	20.42	21.17	6.8 h	68.2 h	682.1 h	137	6.5 h	65.4 h
TARGETID_39633396137198490	20.51	21.27	8.1 h	81.1 h	811.5 h	163	7.8 h	77.8 h
TARGETID_39633432401152732	20.55	21.30	8.6 h	85.5 h	855.0 h	172	8.2 h	81.9 h
TARGETID_39633325958105931	20.83	21.58	14.0 h	139.7 h	1397.3 h	280	13.4 h	133.7 h
TARGETID_39633162619324602	20.85	21.60	14.5 h	144.7 h	1447.5 h	290	13.9 h	138.5 h
TARGETID_39633387157195822	21.06	21.81	21.0 h	210.0 h	2100.1 h	421	20.1 h	200.9 h
TARGETID_39632946793025886	21.26	22.01	30.0 h	300.0 h	3000.2 h	601	28.7 h	287.1 h
TARGETID_39633311366122701	21.27	22.03	31.1 h	310.9 h	3109.5 h	622	29.8 h	297.5 h
J140658.28+161142.6	21.52	22.27	47.8 h	478.4 h	4783.5 h	957	45.8 h	457.9 h
TARGETID_39628379145637914	21.62	22.38	58.3 h	583.2 h	5831.9 h	1167	55.8 h	558.4 h
TARGETID_39628346081936841	21.73	22.49	71.1 h	711.3 h	7113.1 h	1423	68.1 h	681.2 h
TARGETID_39633085267968176	22.00	22.76	116.0 h	1159.9 h	11598.8 h	2320	75.8 h	757.9 h
J143404.58+093528.5	23.00	23.75	705.7 h	7056.9 h	70569.4 h	14114	454.3 h	4543.2 h